Partial transfer of cell-resolved CA1 remapping directions across boundary and reward changes

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Abstract

Hippocampal activity is structured relative to boundaries and rewards, but it is unclear whether the registered-cell direction of change estimated at one coordinate predicts a related change elsewhere. We reanalysed two public longitudinal CA1 calcium-imaging datasets. In an internal-wall arena (seven mice), a wall-minus-open source pattern predicted a non-overlapping wall response 25 cm away (mean Spearman $\overline{\rho} = 0.148$; seven of seven animals positive). The query-matched exact-location prediction was stronger ($\overline{\rho} = 0.230$; shifted-minus-exact $= -0.082$; one of seven animals favoured the shifted predictor). Against all admissible sources at the same 25 cm distance, the correct wall relation had mean animal-level percentile rank 0.612 (seven of seven above chance; descriptive exact tail 1/128), but did not reliably beat the best alternative, and the strict tangential subset remained heterogeneous. Measured-covariate-adjusted cross-location prediction remained positive ($\overline{\rho} = 0.119$; seven of seven), whereas a same-session mirror-open control was estimable in only five animals. In a virtual-track reward task, repeated 120 cm transition classes produced more aligned registered-cell update fields than the 240 cm return transition ($S = 0.0481$; eight of eight confirmatory mice; exact $p = 0.003906$). After non-linear out-of-fold adjustment for speed and licking, the statistic was $S = 0.0238$ (seven of eight; $p = 0.0117$), with only a 1.03% improvement in total held-out trial mean-squared error. CA1 remapping therefore shows partial cell-resolved transfer across repeated spatial relations that is weaker than exact-location recurrence, spatially constrained, and partly coupled to speed and licking.

Keywords: hippocampus; CA1; place cells; boundaries; reward; population code; partial transfer; remapping; cognitive map

1 Introduction

Place fields are organized by environmental geometry and often occur at characteristic distances and directions from boundaries (O'Keefe and Burgess, 1996; Hartley et al., 2000; Barry et al., 2006; Lever et al., 2009). Inserted barriers can produce local fields, duplicate fields, and reshape excitatory and inhibitory boundary responses (Rivard et al., 2004; Wang et al., 2020; Muessig et al., 2024). Reward likewise organizes a subset of hippocampal responses, and place fields may remain tied to the track, shift with the goal, or occupy intermediate reference frames (Gauthier and Tank, 2018; Qian et al., 2025; Sosa et al., 2025a; Yaghoubi et al., 2026). Relative coding of boundaries and rewards is therefore established background, not the phenomenon introduced here.

The narrower question concerns changes rather than static representations. For the same longitudinally registered cells, a boundary insertion or reward relocation defines a direction: which cells increase, which decrease, and where those changes occur. A shared reference frame can generate aligned directions, but it does not guarantee that a direction estimated at one absolute coordinate predicts activity at another. Conversely, aligned directions need not imply position-independent portability; their strength may decline when moved. Related population-geometric questions have revealed parallel coding directions and abstract variables in hippocampal tasks (Bernardi et al., 2020; Courellis et al., 2024). Here longitudinal identity permits a cell-resolved test across physical coordinates.

We reanalysed two public longitudinal dorsal-CA1 datasets. The first followed seven mice through repeated layouts built from internal walls on a common lattice (Lee et al., 2025). We estimated a wall-minus-open contrast at one seam and tested later activity at the same seam or at a non-overlapping seam. The second followed reward switches on a circular virtual track (Sosa et al., 2025a). Within each visual environment, two 120 cm reward moves shared a signed displacement but began and ended at different absolute locations; a 240 cm return move provided a distinct comparison. We froze the reward endpoint after a separate development mouse and tested it in eight confirmatory animals.

The possible outcomes were graded: no cross-location prediction, partial prediction weaker than exact-location recurrence, or position-independent prediction. The wall data supported the middle outcome. The reward analysis provided conceptual convergence, but measured speed and licking carried still stronger transition-class structure and attenuated the neural statistic. Together, the analyses test whether a cell-resolved remapping direction crosses locations and make its spatial limits and coupling to measured behavior part of the result.

2 Results

2.1 Two tests of cell-resolved cross-location transfer

The two datasets differ in task, geometry, and estimator, so their effect sizes were not pooled. They implement the same logical test (Fig. 1). First, a change vector is estimated while preserving registered-cell identity. Second, the same spatial relation is found at another absolute coordinate. Third, alignment with the later change is compared with a matched alternative. In the wall task the change is wall minus open and the alternative is an exact- location or spatial control. In the reward task the change is post minus pre; the two repeated 120 cm transition classes are compared with a 240 cm return.

2.2 A held-out test of cell-resolved wall contrasts

Each candidate wall location was represented as an oriented seam between two adjacent lattice tiles. For each registered cell, environment, and seam, we estimated activity in a narrow local strip and subtracted the corresponding estimate from a bracketing familiar-square session. This square residual reduced stable place preference while retaining the environment-specific local change.

Training geometries supplied two cell-wise profiles at each seam: a mean residual when the seam was blocked by a wall and a mean residual when it was open. Their difference was the estimated wall-related contrast. The neural outcome from the later target geometry was never used to form that source contrast. Prediction was scored by a rank correlation across held-out registered cells. We report animal-level means and sign counts rather than treating cells or queries as independent biological replicates.

2.3 A local wall-open contrast recurs across global geometries

For exact-location recurrence, the broader wall context was matched between training and held-out comparisons. Across 252 held-out predictions, the wall-derived profile correlated more strongly with later wall responses than did the open-derived profile. The mean wall-minus-open correlation advantage was 0.184 and was positive in all seven animals (Fig. 2A). The sign was also positive in all animals when horizontal and vertical seams were evaluated separately.

This is more specific than a population-average increase near walls. The predictor is indexed by registered cell identity: cells with positive or negative values in the estimated local contrast tended to retain those signs when the same seam appeared in another global geometry. Because the seam was unchanged, this result addresses recurrence at one location, not prediction across locations.

2.4 A wall-related population contrast predicts a distinct location

For each eligible target, a wall-minus-open vector was estimated at a distinct source seam with the same signed wall normal. The source and target seams were 25 cm apart, their local analysis bins did not overlap, and source training sessions were disjoint from the target session. The nearest source-target bins were 5 cm apart for tangential translations and 15 cm apart for normal-axis translations. The source pattern was constructed from wall and open training states across multiple geometries.

The shifted source vector predicted the later target-wall residual with mean correlation 0.148, positive in all seven animals (Fig. 2B). It also outperformed a source-open predictor by 0.152, and was more aligned with the target-wall response than with a training-derived target-open profile by 0.216, again positive in every animal. This cell-by-cell result is more specific than elevated activity at two wall locations: the signed contrast estimated at one coordinate predicted the response at another.

The exact-location benchmark was higher. On the same queries, cells, and support, it predicted the target response with mean correlation 0.230. Cross-location prediction was lower by 0.082, and only one of seven animals favored the shifted predictor. Thus part of the cell-resolved pattern crossed locations, but a stronger component remained tied to its original coordinate.

2.5 Measured-covariate adjustment and spatial controls

Because animals may run differently near internal barriers, we re-estimated local neural response maps with covariates for speed, first and second circular harmonics of heading, elapsed time, and spatial sampling. The covariate-adjusted cross-location prediction remained positive in all seven animals ($\overline{\rho} = 0.119$). It correlated 0.172 more strongly with the target wall than with a training-derived target-open profile, and it exceeded a source-open predictor by 0.138; both differences were positive in all seven animals. Omitting elapsed time changed these values by less than 0.003.

A frozen-local empirical baseline then replaced the single selected spatial control with full enumeration. For every target, we scored all non-target source seams that met the precommitted exact 25 cm distance, wall/open training, non-overlap, common-cell, common-bin, and occupancy requirements. Sources were selected by geome-

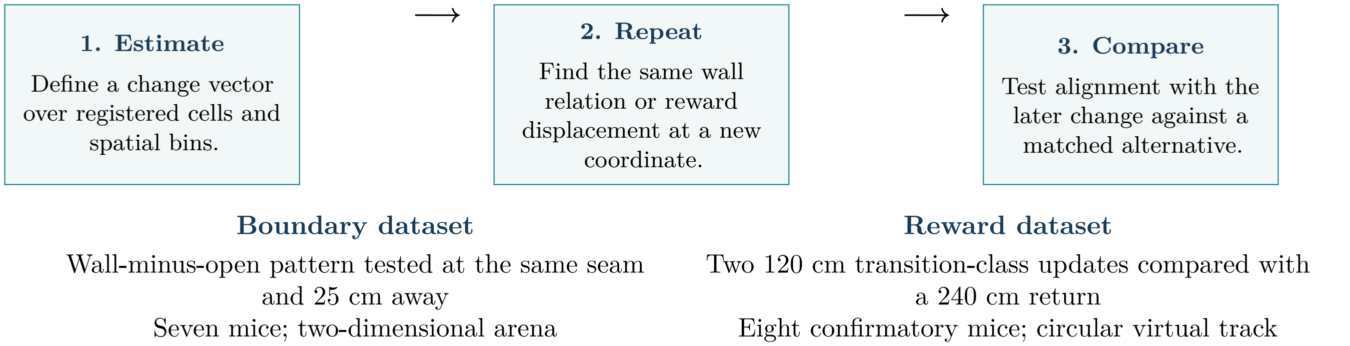

Figure 1: Common analysis logic across two independent datasets. Both analyses ask whether the direction of a registered-cell population change recurs when a spatial relation reappears at another absolute location. The estimators and numerical scales differ and are analysed separately.

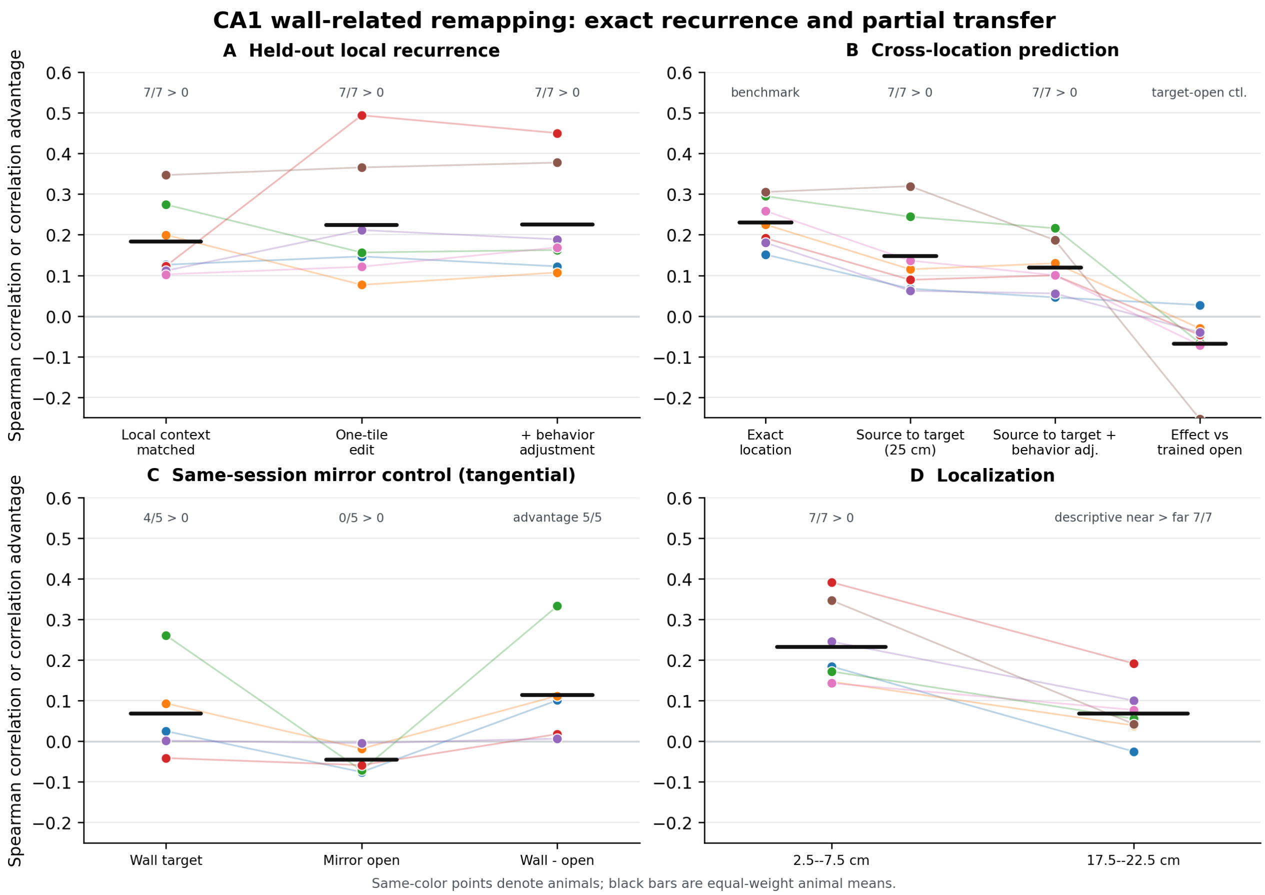

Figure 2: Exact recurrence, cross-location prediction, and key controls. **A**, same-seam wall-minus-open prediction under three estimators. **B**, matched exact-location and source-to-target predictions, including adjustment for measured running covariates and the target-open comparison. **C**, wall and mirror-open targets within the same session; five animals had paired support. **D**, descriptive near and far bands with separate eligibility. Points are animal means; bars are across-animal means.

try and support, never by target neural similarity. The score for the predicted wall-normal relation was ranked among all admissible source scores and averaged within exposure pair and animal before cohort summaries.

The mean animal-level percentile rank was 0.612, above the chance value of 0.5 in all seven animals (descriptive exact sign-flip tail 1/128; Fig. 3B). The correct relation exceeded the mean alternative correlation by 0.102, again positive in seven of seven animals. It exceeded the best alternative by 0.059, but this comparison was positive in only five of seven (tail fraction 0.078; Fig. 3C). Raw-rate scoring reproduced the rank and mean-alternative results. Thus the effect does not depend on choosing one convenient wrong orientation, but the correct relation is not uniquely strongest for every animal.

The fully enumerated control does not eliminate spatial continuity. Restricting the candidates to tangential translations also matches the 5 cm nearest-strip lag, but leaves only six animals and as few as three queries in one animal. The mean percentile rank was 0.571, positive in four of six animals (tail fraction 0.078); the correct-minus-mean-alternative correlation was 0.040, positive in three of six (tail fraction 0.25; Fig. 3D). The most tightly distance-matched stratum therefore remains heterogeneous.

A stronger fitted-model comparison was also frozen before target scoring but could not be estimated under its leakage barrier. The combined model required its empirical-update weight to be calibrated on an exposure pair sharing no neural session with the outer test pair. Six animals had only three exposure cycles, and one had two, so no valid calibration set existed for every animal and outer pair. No partial fitted-model endpoint was inspected. Consequently, the enumeration addresses selected-control bias but does not show that the source pattern adds information beyond a fitted boundary-relative or smooth place-field model.

Generic nearby place-field structure could still be shared by the source and target strips. A stricter within-session control was specified before its neural endpoint was inspected. For each wall target, a mirror-open seam in the same target session was matched for translation distance, axis, signed normal, relative bins, and registered cells. The geometry permits this match only for tangential translations. Although 104 labeled configurations were feasible, paired-support gates retained 72 triplets in five animals; two animals had no admissible control.

On this restricted subset, the globally demeaned source vector favored the wall target over the mirror-open target by 0.114, positive in all five eligible animals (Fig. 2C). Absolute source-to-wall correlation was positive in four of five animals ($\bar{\rho} = 0.068$), and a direct within-session wall-minus-open difference-vector score was positive in four of five ($\bar{\rho} = 0.077$). The raw-rate primary was positive in only three of five. The exact animal-level sign-flip tail fraction for the demeaned advantage was 1/32, and every leave-one-animal-out mean remained positive. In the supported demeaned tangential cases, the source contrast distinguished a wall target from an equally displaced open location. This control does not cover all seven animals, normal-axis translations, or raw-rate specificity.

A coherent source-cell identity permutation further showed descriptive upper-tail fractions below 0.01 in five animals and values of 0.069 and 0.060 in the other two. Registered-cell correspondence therefore contributes to the prediction, although the effect is not uniformly strong. Additional exploratory analyses of direction, composition, experience, and a former-barrier endpoint are reported in Supplementary Section 5 rather than treated as tests of the central phenomenon.

Across animal-level comparisons, cross-location prediction was consistently positive but lower than exact-location prediction. The correct relation ranked above the full exact-distance candidate set, and the restricted mirror-open comparison favored wall targets, but the strict tangential match remained heterogeneous (Fig. 4).

2.6 Repeated reward-transition classes align CA1 remapping directions

The reward dataset provides an independent test in a different geometry. Each confirmatory mouse contributed two three-day sequences, one in each visual environment. Within a sequence, two transitions moved the hidden reward 120 cm in the same signed direction, but from different starting positions. The third transition moved it 240 cm in the opposite direction. We aligned each post-minus-pre population update to the old reward and retained cells registered across all three days.

For each mouse and environment, we measured the Pearson correlation between the two short update fields and subtracted the mean of their correlations with the long return update:

$$S = r(\Delta_{120}^{(1)}, \Delta_{120}^{(2)}) - \frac{r(\Delta_{120}^{(1)}, \Delta_{240}) + r(\Delta_{240}, \Delta_{120}^{(2)})}{2}. \quad (1)$$

The mean same-displacement correlation was 0.120, compared with 0.071 and 0.073 for the two short–long pairs. Their difference was $S = 0.0481$ (mouse-bootstrap 95% interval $[0.0234, 0.0757]$), positive in all eight confirmatory mice (exact one-sided sign-flip $p = 0.003906$). Fourteen of the 16 mouse–environment values were positive. Deconvolved events gave $S = 0.0401$, again positive in eight of eight mice ($p = 0.003906$). Thus the two 120 cm transitions produced preferentially aligned update directions after their absolute start and end positions changed.

The comparison identifies the relation represented in the available design: the control move differs in both sign and distance. It therefore does not separate displacement direction from step length. The analysis tests the direction of the population update; amplitude portability was not tested.

2.7 Reward-transition specificity survives drift, null, and parameter tests

A pseudo-update constructed before each reward switch used the same mice, days, environments, registered cells, and transition schedule. Its mean specificity was 0.00235, with four of eight mouse means positive ($p = 0.359$); the corresponding post-switch drift estimate was -0.00114 ($p = 0.578$). The three fixed-condition mice lacked public cross-day ROI identities, so they could not support the same registered-cell endpoint. Within each of their 18 available sessions, however, the direction of early drift did not recur later: the mean quarter-to-quarter correlation across the three mice was -0.00352 (Fig. 5D).

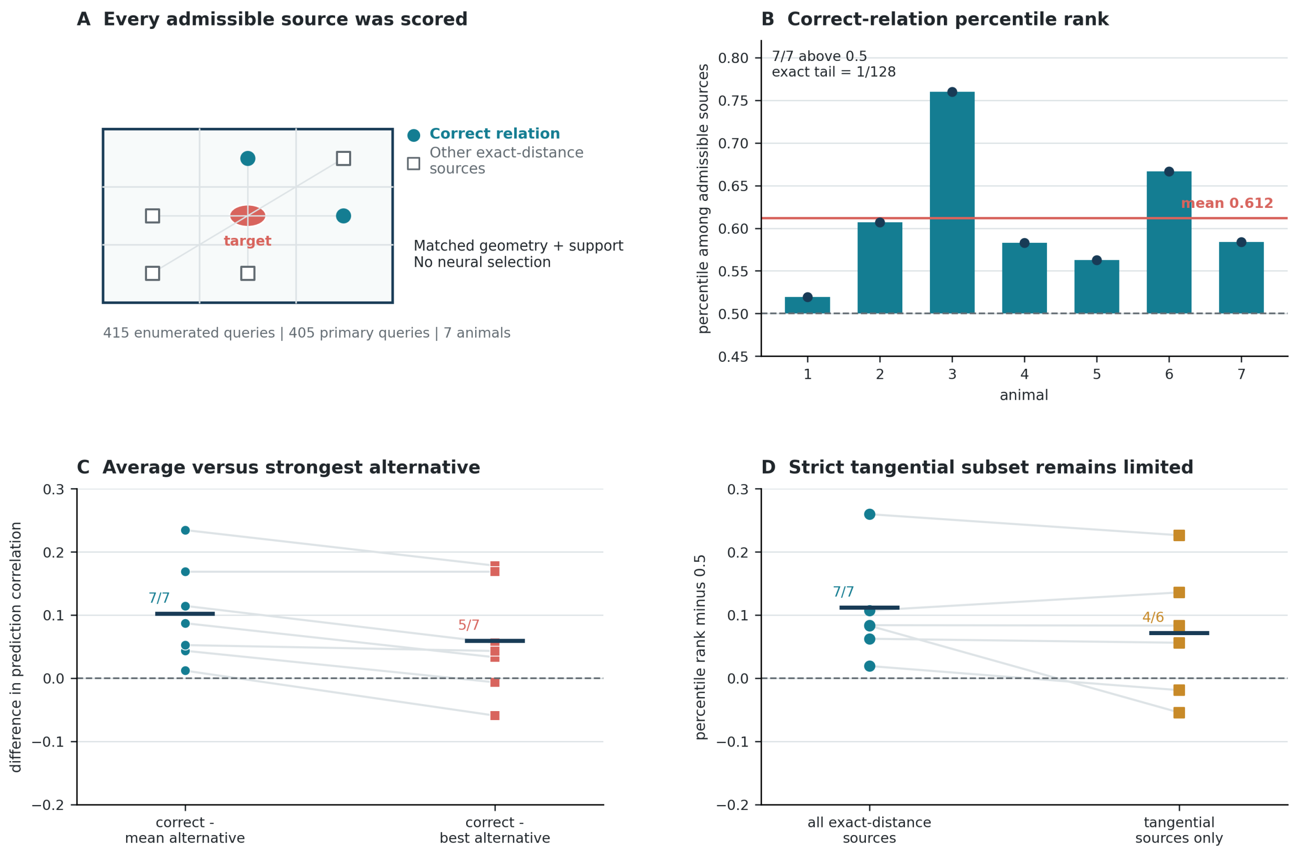

Figure 3: The correct wall relation against fully enumerated empirical spatial alternatives. **A**, schematic of the matching logic; positions are illustrative rather than measured seam coordinates. Every admissible source at the exact target-midpoint distance was retained without neural selection. **B**, animal-level percentile rank of the correct-relation score among all admissible source scores. **C**, correct-relation score minus the mean or best alternative. **D**, centered percentile rank for all exact-distance sources and the support-limited tangential subset. Lines in C and D connect animal estimates; horizontal bars mark cohort means. Exact tails are descriptive animal-level sign-flip calibrations.

Two label-preserving nulls also separated the reward result from chance. A trial-membership null had mean 0.00010 and 95% interval $[-0.01161, 0.01308]$; none of 199 frozen draws reached the observed value (plus-one $p = 0.005$). A 200,000-draw repeated-transition-label null had mean -0.000006, 95% interval $[-0.02570, 0.02775]$, and $p = 0.000055$ (Fig. 5B).

All named normalization and spatial-resolution variants retained a positive cohort mean (range 0.0431–0.0679). All 500 deterministic half-cell subsamples were positive (mean 0.0488, descriptive 95% interval $[0.0379, 0.0610]$), every leave-one-mouse-out mean was positive, and all 27 registration-threshold combinations retained the sign. These checks address drift, arbitrary labels, cell sampling, preprocessing, and registration thresholds without changing the frozen biological unit.

2.8 Reward-transition geometry is shared with speed and licking

After post-switch speed and licking were translated into the neural old-reward frame, both behaviors carried strong displacement-specific structure. The speed-only endpoint was $S = 0.2261$ and the lick-only endpoint was $S = 0.3246$; both were positive in eight of eight mice ($p = 0.003906$). A spatial aggregate linear adjustment left a positive dF/F mean ($S = 0.0253$) but only five of eight mouse means were positive ($p = 0.0547$), below its frozen six-of-eight consistency requirement. The neural result therefore cannot be described as unrelated to how the animals ran and licked.

We then conditioned neural activity on speed and licking trial by trial. At each of 45 positions, a Gaussian radial-basis kernel used speed and $\log(1 + \text{lick})$ in the adjacent and current bins, whole-trial speed and licking, and missingness indicators to predict all registered cells. Five-fold cross-fitting held out consecutive trial blocks within both pre- and post-switch conditions; all feature scaling, kernel bandwidths, cell means, and regression weights were estimated without the held-out trials. Reward condition and displacement were not model inputs. We recomputed Eq. 1 from the out-of-fold residual activity.

At the frozen primary regularization ($\lambda = 1$), residual dF/F specificity was $S = 0.0238$, positive in seven of

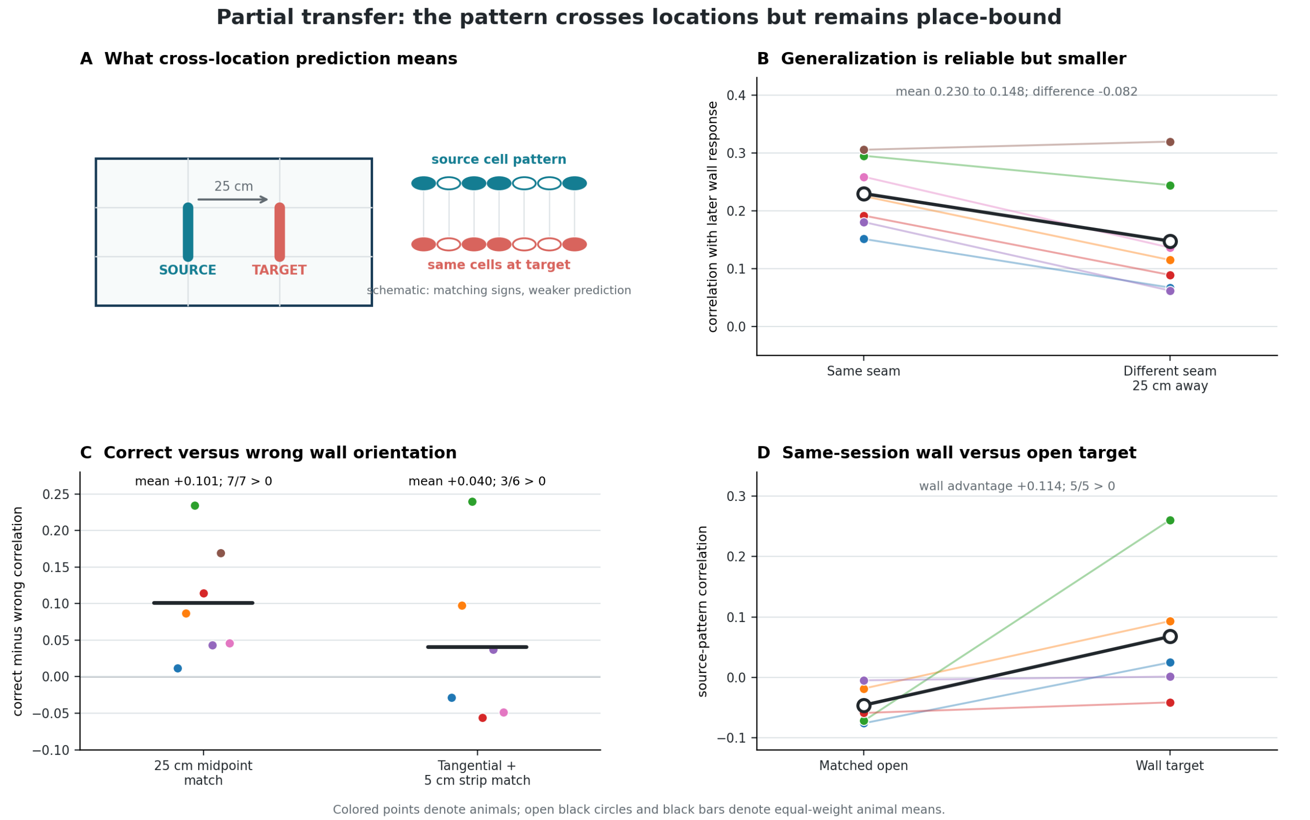

Figure 4: The cell-resolved pattern crosses locations but loses predictive strength. **A**, schematic of a source wall-related cell pattern tested at a target seam 25 cm away. Filled and open circles denote schematic signed cell responses, not additional measurements. **B**, paired exact-location and cross-location correlations in seven animals. **C**, correct-minus-wrong orientation effects for the pooled 25 cm control and the strictly matched tangential subset. **D**, source-pattern correlations with same-session mirror-open and wall targets in five eligible animals. Quantitative panels show globally demeaned animal-level estimates.

eight mice ($p = 0.0117$; Fig. 6A). Residual events gave $S = 0.0153$, also positive in seven of eight mice ($p = 0.0195$). The dF/F mean was 49.5% of the original, and every mouse showed attenuation. Fixed $\lambda = 0.1$ and 10 sensitivities also retained positive dF/F means; the event result was positive at all three values and met the six-mouse consistency threshold at $\lambda = 0.1$.

Speed and licking explained a modest fraction of total held-out trial variance. At $\lambda = 1$, pooled mean-squared error improved by 1.03% for dF/F and 0.65% for events relative to a position-only cell mean. The more regularized $\lambda = 10$ model improved prediction by 2.77% and 2.56%, respectively, while removing less of the specificity signal. The flexible $\lambda = 0.1$ model overfit and performed worse than the position-only predictor, yet its dF/F residual specificity remained positive. The primary result therefore does not depend on selecting the model that yields the most favorable neural endpoint.

2.9 Convergence and non-equivalence across paradigms

The wall and reward analyses are not replications of an identical statistic. One predicts a local wall-related response across seams; the other compares full cell-by-position update fields across reward transitions. Their convergence lies in the registered-cell relation: a change direction estimated at one absolute location predicts a related change elsewhere, in seven of seven and eight of eight primary-animal summaries, respectively (Table 1). The differences are equally informative. Wall prediction declined across physical distance, and out-of-fold conditioning on speed and licking attenuated about half of the reward-specificity statistic. Their conceptual intersection is partial transfer under a repeated spatial relation, not position-independent portability across locations or tasks.

3 Discussion

Across two longitudinal datasets, a registered-cell direction estimated at one absolute location predicted a related CA1 change elsewhere. What transferred was a correlation structure over the identities of cells that rose and fell; cross-location prediction was weaker than same-location recurrence, and amplitude portability was not tested. In the wall task, cross-location prediction was

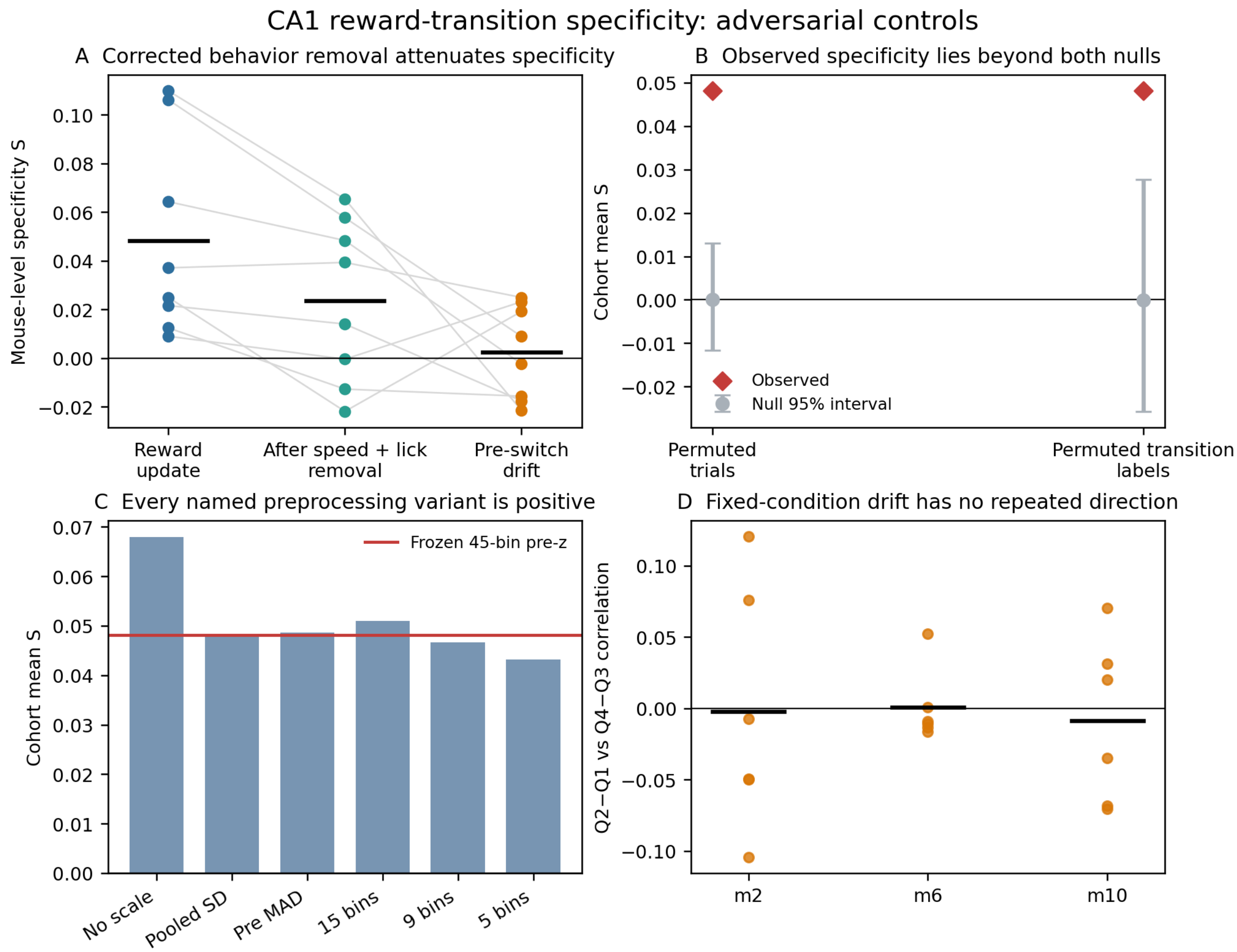

Figure 5: Controls for reward-transition specificity. **A**, the original registered-cell endpoint, a corrected aggregate speed-and-lick residual, and pre-switch drift; points are mouse means. **B**, the observed cohort value and 95% intervals from trial-membership and transition-label nulls. **C**, named normalization and spatial-binning variants. **D**, within-session repeated-drift correlations in three fixed-condition mice. The aggregate speed-and-lick residual in A did not meet its predeclared cross-mouse consistency gate; trial-level conditioning is shown in Fig. 6.

0.148 compared with 0.230 on the same target support at the exact seam. Registered identity is essential to this statement: an average increase near two walls would not predict which particular cells change together. The reward result provided independent conceptual convergence, while also showing that transition-class geometry is substantially shared with speed and licking. The common result is therefore partial cell-resolved transfer, not position-independent portability or independence from measured behavior.

Stable relative tuning is a parsimonious candidate explanation for the wall result. Boundary-vector and place-field models predict responses from distance, direction, and environmental geometry (Hartley et al., 2000; Barry et al., 2006; Muessig et al., 2024); the source study further showed that local boundary features predict representational similarity in these same recordings (Lee et al., 2025). Subtracting open from wall states can turn such stable tuning into a reproducible population difference vector. The present contribution is not an additional boundary code, but a longitudinal measurement of that difference: the source pattern predicts a distinct seam, and the exact-versus-shifted comparison measures how much prediction is lost under translation.

The fully enumerated baseline sharpens this descriptive result. The correct wall relation ranked above all exact-distance admissible sources on average in every animal and exceeded the mean alternative in every animal. It did not reliably exceed the best alternative, and the strict tangential subset was heterogeneous. The precommitted fitted boundary-model comparison could not be calibrated without neural-session leakage because too few independent exposure cycles were available. Thus selected-control bias is less plausible, but a boundary-relative or smooth-place model may still exhaust the observed cross-location alignment. Additional independently usable exposure cycles or a separate development cohort are needed to distinguish these possibilities.

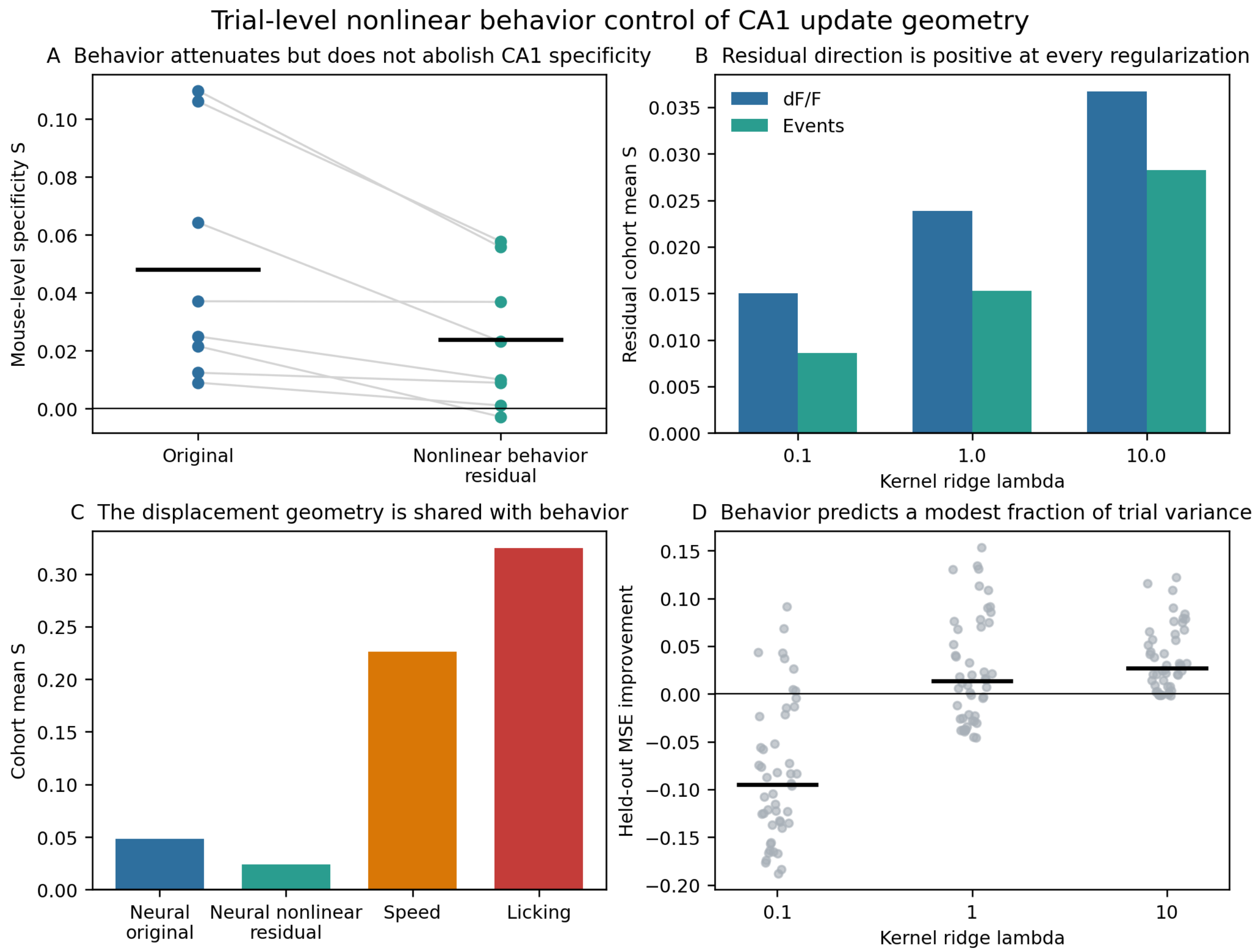

Figure 6: Trial-level relationship between speed, licking, and CA1 reward updates. **A**, mouse-level specificity before and after nonlinear out-of-fold conditioning on speed and licking. **B**, residual dF/F and event specificity across the three fixed regularization values. **C**, the original neural endpoint, the primary nonlinear residual, and speed/lick-only endpoints. **D**, held-out neural prediction improvement for each session and regularization value; black bars show pooled values.

The reward result adds a temporally separated test. Reward-linked cells and mixtures of track- and goal-referenced fields are established (Gauthier and Tank, 2018; Qian et al., 2025; Sosa et al., 2025a), and predictive models can generate learning-dependent reward representations (Yaghoubi et al., 2026). Here, full registered-cell update fields from the two 120 cm transitions were preferentially aligned. Registered pre-switch changes, fixed-session drift, permuted trials, and permuted transition labels did not reproduce the effect, and it was stable across declared preprocessing, cell-sampling, and registration sensitivities. The design nevertheless confounds displacement sign with distance, so the result identifies a repeated transition class without isolating displacement direction from step length.

Speed and licking are part of this reward-related geometry. They had larger specificity values than CA1, and nonlinear out-of-fold conditioning attenuated the neural specificity statistic in every mouse. A smaller dF/F and event relation remained, but the primary speed-and-lick model improved total held-out trial mean-squared error by only 1.03% and 0.65%, respectively. Accordingly, “about half” refers to attenuation of the chosen dF/F specificity statistic, not to variance in CA1 activity. Running and licking may be causes, consequences, or parallel readouts of a latent task variable; pupil, posture, acceleration, consumption, and engagement were not measured jointly.

Several limits remain dataset-specific. The mirror-open wall control covered five animals and only tangential translations, the wall sequence was fixed, and the analyses select cells stable enough to register longitudinally. The reward fixed-condition animals lack public cross-day identities. The wall and reward statistics also differ in scale and estimand; they are not pooled numerical replications. These restrictions preserve the distinction among same-location recurrence, partial cross-location transfer, and position-independent portability.

A prospective test should counterbalance wall and goal changes across absolute locations while interleaving no-change, opposite-direction, and same-distance open conditions. Additional exposure cycles would permit leakage-

Table 1: **Evidence and limits for partial transfer of CA1 remapping directions.** Values are means across animal-level estimates within each dataset. Numerical scales differ across estimators and are not pooled.

Dataset	Test	Mean	Sign	Supported interpretation
Wall	Context-matched exact seam	0.184	7/7	Local wall-open population contrast recurs across layouts.
Wall	Cross-location source effect	0.148	7/7	A cell-resolved wall contrast predicts a different location.
Wall	Query-matched exact effect	0.230	7/7	Same-location prediction is stronger.
Wall	Exact-distance source rank	0.612	7/7	Correct relation ranks above chance across all admissible sources.
Wall	Covariate-adjusted cross-location effect	0.119	7/7	The cross-location result remains after adjustment for measured covariates.
Wall	Mirror-open advantage	0.114	5/5	Demeaned tangential subset favors a wall over an equally displaced open seam.
Reward	Original dF/F specificity	0.0481	8/8	Repeated 120 cm transitions align more than either aligns with the return.
Reward	Deconvolved-event specificity	0.0401	8/8	The result is preserved in an independently processed signal.
Reward	Nonlinear dF/F residual	0.0238	7/8	A smaller CA1 component remains after out-of-fold speed and lick conditioning.
Reward	Speed-only / lick-only	0.226/0.325	8/8	The displacement relation strongly structures speed and licking.

free model calibration, and simultaneous movement, licking, eye, and posture measurements would separate neural residuals from measured movements and licking. The present estimates provide power-analysis targets without equating the two dataset-specific effects.

4 Materials and methods

4.1 Dataset description

4.1.1 Boundary-layout dataset

We reanalysed the public longitudinal dorsal-CA1 calcium-imaging dataset of Lee et al. (2025, 2024). It contains seven mice, 5,413 longitudinal cell identities, and 207 recording sessions. Each mouse foraged in the same familiar square outer arena and in nine internal-wall layouts defined on a common spatial lattice. The released data include position and sampling, spatial rate maps, geometry labels, and cell-registration identities across sessions. Six mice completed three exposure cycles (31 sessions each), and one completed two (21 sessions). Within every cycle, the nine wall layouts occurred in a fixed order between familiar-square sessions. Consequently, environment and sequence position were not independently varied. The original study describes surgery, imaging, behavioral procedures, preprocessing, and cell registration; no new animal experiments were performed here.

4.1.2 Reward-relocation dataset

We independently reanalysed the public dorsal-CA1 two-photon dataset of Sosa et al. (2025a,c). Mice ran on a 450 cm circular virtual track with three possible unmarked reward zones and two visual environments. The processed release contains trial-by-position dF/F, deconvolved events, speed, licking, reward metadata, and published cell annotations. Imaging masks and ROI indices used to reproduce cross-day cell matching came from DANDI 001361 (Sosa et al., 2025b); speed and lick variables came from the processed release because the audited NWB speed and lick arrays were not used.

Eleven mice performed reward switches and three remained in fixed conditions. Mouse 11 defined the endpoint in a development pilot and was excluded from confirmation. A metadata-only registration audit excluded mice 17 and 18 because curated NWB cell order could not be mapped exactly to the processed activity axes. The confirmatory cohort therefore comprised mice 3, 4, 7, 12, 13, 14, 15, and 19. Each provided two within-environment three-day sequences (days 3, 5, 7 and days 10, 12, 14). Day 8 was excluded because reward and visual environment changed together. Primary cell selection used only cross-day registration and training-only measurement reliability; published reward-relative labels did not select cells.

4.2 Boundary representation and primary estimators

The arena was expressed in a common allocentric coordinate system. A candidate internal-wall location was represented as an oriented seam $s = (p \to q)$ between adjacent lattice tiles, with a signed normal from tile p to tile q. The binary state $z_{e,s}$ indicated whether environment e blocked that adjacency. Local strips were partitioned into 5 cm spatial bins at defined distances from the seam. Analyses used only bins and registered cells satisfying the relevant occupancy, activity, and common-support gates. Source and target strips in the cross-location analysis shared no 5 cm bin.

Local response vectors. For registered cell i, environment e, and seam s, let $r_{i,e,s}$ be activity averaged over the eligible local bins. Let $b(e)$ denote the paired familiar-square session. We subtract that square baseline and collect the changes across the n shared cells:

$$\begin{aligned}\delta_{i,e,s} &= r_{i,e,s} - r_{i,b(e),s},\\ \boldsymbol{\delta}_{e,s} &= \left(\delta_{1,e,s}, \dots, \delta_{n,e,s}\right)^{\mathsf{T}}.\end{aligned} \tag{2}$$

For globally demeaned analyses, the vector mean was removed before correlation. The resulting score depends on relative cell-specific changes rather than a population-wide rate shift.

Training profiles and held-out scoring. For a target query (e^*, s^*), training environments were selected without using the neural response of e^*. At a source seam s, the wall and open profiles were

$$\begin{aligned}\boldsymbol{\mu}_{W,s} &= \mathrm{mean}_{e\in\mathcal{T}:z_{e,s}=1}\, \boldsymbol{\delta}_{e,s},\\ \boldsymbol{\mu}_{O,s} &= \mathrm{mean}_{e\in\mathcal{T}:z_{e,s}=0}\, \boldsymbol{\delta}_{e,s}.\end{aligned}$$

computed on the common registered-cell support. The estimated source effect was $\mathbf{d}_s = \boldsymbol{\mu}_{W,s} - \boldsymbol{\mu}_{O,s}$. The held-out outcome was $\boldsymbol{\delta}_{e^*,s^*}$. Scores were Spearman correlations across registered cells. Wall-open advantages compare correlations within the same query and cell support.

The exact-location estimator set $s = s^*$. Its context-matched version restricted training comparisons so the broader internal-wall context was balanced between the wall and open profiles. The reported exact-location benchmark for cross-location prediction was recomputed on the exact queries, sessions, cells, and local supports eligible for the shifted analysis.

Single-tile counterfactual. The u and o geometries differed at one lattice tile: u blocked tiles $\{4, 5\}$, whereas o blocked only tile $\{4\}$. We formed the cell-wise $u - o$ contrast at the focal seam and evaluated wall and open outcomes at that seam in held-out i, l, and t geometries. Near (2.5–7.5 cm) and far (17.5–22.5 cm) bands were screened separately, so their difference is descriptive. For the order-placebo analysis, we formed later-minus-earlier contrasts at unchanged seams with the same processing and compared their signs with the actual $u - o$ contrast.

Cross-location prediction. For each eligible target seam, the source seam was a non-overlapping lattice seam 25 cm away with the same signed wall normal and admissible wall/open training support. Source training and target outcome sessions were disjoint. The primary shifted score was

$$\rho_{\mathrm{shift}} = \rho_{\mathrm{S}}\left(\mathbf{d}_{s_{\mathrm{source}}}, \boldsymbol{\delta}_{e^*,s_{\mathrm{target}}}\right).$$

Secondary specificity scores compared the source effect with a training-derived open profile at the target and compared source-wall with source-open predictors. The exact-location benchmark used $\mathbf{d}_{s_{\mathrm{target}}}$ on the identical held-out query support. Stored-map and covariate-adjusted audits agreed on 405 target queries and 658 source pairs, including their session and seam identities and their counts of common cells and bins. That audit did not hash the complete cell-ID or bin-coordinate arrays.

Measured-covariate-adjusted response estimates. We estimated activity jointly with spatial-bin indicators and nuisance covariates for instantaneous speed, sine and cosine of heading, second heading harmonics, and elapsed time. Spatial effects were evaluated at a common reference covariate distribution calibrated from the first familiar-square session. The no-time sensitivity omitted elapsed time but retained the other covariates. Target-session activity was used to estimate nuisance coefficients for that target map, but target neural outcomes never entered source-profile training.

4.3 Boundary controls and adjudication

Same-session mirror-open control. The mirror-open seam was specified before its neural endpoint was inspected. For each target wall, it lay in the same target session and matched the source-wall pair in translation distance, translation axis, signed normal, relative spatial bins, and registered cells. Both source-to-target nearest-bin distances were 5 cm. Only tangential translations admitted this exact geometric pairing. Primary scoring used globally demeaned source and target vectors. Eligibility required a complete source, wall-target, and open-target triplet on common support.

Spatial-null calibration. We calibrated two empirical spatial controls at the animal level. The first compared correctly oriented source seams with incorrectly oriented sources at the same 25 cm target-midpoint distance. A tangential stratum additionally matched translation axis and the 5 cm nearest-strip lag. The second compared the wall target with the same-session reflected-open target described above. Both controls retain the observed neural maps, including their place-field smoothness. For each paired animal-level contrast, we enumerated all 2^N sign assignments and calculated the fraction of null means at least as large as the observed mean. These are descriptive exact sign-flip calibrations, not confirmatory population p-values.

Fully enumerated empirical spatial baseline. The empirical baseline was frozen in a committed protocol before its target neural endpoint was inspected. It inherited the 405 target queries and 658 correct-relation source pairs in the audited cross-location ledger, while also enumerating geometry- and support-admissible alternatives. The source ledger's SHA-256 digest and expected counts were fixed in advance. Each candidate source had an exact 25 cm target-midpoint distance, wall and open training states, non-overlapping target support, disjoint source-training and target neural sessions, at least 20 common cells, at least six common bins, and at least 0.5 s occupancy per common bin. Target neural similarity did not enter source eligibility.

Within a query, every candidate source produced the same Spearman prediction score used in the primary shifted analysis. If more than one source had the predicted signed wall-normal relation, their scores were averaged with equal weight. The primary endpoint was the empirical midrank of this relation mean among all finite candidate-source scores, minus the chance rank of 0.5. Secondary endpoints were the correct-relation mean minus the mean alternative, minus the best alternative, and the centered indicator that the correct mean beat every alternative. Queries were averaged within exposure pair, exposure pairs within animal, and animals within cohort. Exact animal-level sign flips were enumerated. The primary rate mode removed each local vector's global mean; raw local rates were a fixed sensitivity. A support-limited tier retained only tangential sources, thereby additionally matching the nearest-strip lag.

The same protocol specified fitted spatial/place, boundary-relative, empirical- update, and combined models. The combined model's empirical-update coefficient had to be calibrated on a training-only pseudoquery that included neither neural session in the outer test pair. No such calibration pair existed for every outer pair and all seven animals: six animals had three exposure cycles and one had two. The predeclared stopping rule was therefore applied before any fitted- model target endpoint was calculated.

Cell-identity permutation. For each animal, we generated 999 coherent random mappings of source-cell identities while preserving query geometry, target vectors, support sizes, and the mapping across all queries in that permutation. The descriptive upper-tail fraction was $(1 + n_{\text{null}\geq\text{observed}})/(1000)$. Correlations and registration constraints limit exchangeability; the fractions were used as identity-specificity diagnostics rather than inferential population p-values.

Direction, composition, and experience analyses. Direction controls compared eligible source-target pairs with same or opposite signed wall normals while documenting translation axis, distance, overlap, and layout matching. No single comparison simultaneously matched all of these features across the full cohort.

For composition, we identified complete wall-state factorials in which two single-wall effects and their joint state could be estimated. The additive prediction was the sum of the two single effects. It was compared with an oracle best-single-component predictor using correlation and normalized prediction error. Only two repetitive factorials and five eligible animals were available.

Experience analyses contrasted transfer changes across exposure blocks using difference-in-differences variants. Because geometry, order, and exposure were not independently randomized, these estimates were treated as exploratory sensitivity analyses and not as evidence for learning.

Independent former-barrier analysis. We used the public neutral-barrier data from Blair et al. (2023b) to test whether the former barrier center retained an excess neural similarity relative to matched control locations after removal. The primary endpoint was a Pearson correlation among active cells, averaged as a former-barrier center-minus-control contrast. Its definition was written before the neural endpoint was run in the local project workflow. Spearman scoring and leave-one-animal-out summaries were sensitivities.

4.4 Reward-transition analysis and controls

Registered-cell update fields. For each reward-switch session, neural activity had dimensions trial by position by cell. We used 45 circular 10 cm bins and cells registered across all three days of a mouse–environment sequence. Each cell was centered and scaled using only its pre-switch samples, after which pre- and post-switch maps were averaged separately and circularly shifted so position zero marked the old reward. For transition k, the registered population update was

$$\boldsymbol{\Delta}_k(x) = \bar{\mathbf{z}}_k^{\text{post}}(x) - \bar{\mathbf{z}}_k^{\text{pre}}(x).$$

Pearson correlations were calculated over common finite cell-by-position entries after flattening the field. The three-day specificity statistic was Eq. 1. It was calculated separately in the two visual environments and averaged within mouse. dF/F was primary; identically processed deconvolved events provided a signal-level robustness analysis.

The primary endpoint was redesigned after a declared mouse-11 development analysis rejected unit-amplitude prediction but identified the correlation- based contrast. The scoring code, eligibility rules, animal-level test, and remaining confirmatory cohort were fixed before neural correlations from the eight confirmatory mice were computed. The mouse bootstrap resampled cells as blocks with all their position bins for a descriptive interval. Confirmatory inference used the eight mouse means in a complete one-sided 2^8 sign-flip test.

Drift, null, and robustness controls. Registered pre-switch drift split each session's pre trials into chronological halves, formed late-minus-early maps, and applied the same three-transition specificity statistic. Post-switch drift was calculated analogously. The processed release does not preserve cross-day identities for fixed-condition mice and DANDI 001361 contains no corresponding ROI assets. We therefore did not infer those identities from activity. The declared contingency split each fixed session into quarters and correlated Q2-minus-Q1 with Q4-minus-Q3; mouse-level means were reported descriptively.

The trial-membership null permuted trials into pseudo pre/post groups while preserving session structure and recomputed the full endpoint for 199 frozen draws. The transition-label null preserved the three observed update fields within each mouse–environment and randomly assigned which pair counted as the repeated transition; 200,000 draws were used. Monte Carlo tails used the plus-one correction. Predeclared sensitivities varied cell scaling, robust scale, and spatial resolution; resampled half the cells 500 times while keeping their spatial bins together; recomputed leave-one-mouse-out means; and applied 27 grids over minimum common-cell count, image correlation, and median ROI intersection-over-union.

Trial-level nonlinear speed-and-lick model. Upstream speed and lick profiles were centered separately on the old and new reward. We reordered their circular bins and translated every post-switch trial by the signed reward displacement (± 12 or ± 24 ten-centimeter bins), placing the neural, speed, and lick arrays in the old-reward frame. Speed- and lick-only updates then used the same three-transition specificity statistic as the neural maps.

The nonlinear model was fit separately at each position. Predictors were speed and $\log(1 + \text{lick rate})$ in the previous, current, and next circular bins; whole-trial means of both quantities; and a missingness indicator for each feature. A Gaussian radial-basis kernel ridge regression predicted all registered cells jointly. The bandwidth squared was the median positive pairwise squared distance among training feature vectors. Five consecutive folds were constructed within the pre and post conditions separately. Feature medians and scales, bandwidth, cell means, and regression weights were learned from four folds and applied to the held-out pre and post blocks. Missing features were replaced by their training median and flagged. Reward condition, displacement, and trial number were excluded from the predictors.

Out-of-fold residuals retained trial, position, and registered-cell identity. We formed residual pre/post maps with the original pre-only neural scaling and recomputed Eq. 1. The frozen primary regularization was $\lambda = 1$; $\lambda = 0.1$ and 10 were fixed sensitivities. Predictive diagnostics compared held-out squared error with a position-specific training- cell mean that used no speed or lick predictors. A separate aggregate linear control regressed each cell's spatial update on corrected speed and lick updates. Its frozen cross-mouse consistency gate was evaluated independently of the nonlinear endpoint.

4.5 Statistical reporting and reproducibility

Animals were treated as the biological units. We report animal-level means, sign counts, query counts, and explicitly labeled descriptive permutation fractions. No cell-level or query-level observation was presented as an independent animal replicate. The wall analyses are exploratory: neither the original experiment nor the reanalysis was preregistered for the cross-location hypothesis. The reward endpoint is confirmatory relative to the declared local workflow: mouse 11 was used for development, and code plus gates were committed before scoring the remaining cohort. This process declaration is not a prospective registration of the original experiment.

All figure values derive from tracked machine-readable source files. The repository contains scripts, eligibility ledgers, result summaries, design locks, test suites, and an inspectable Git history. Accepted reward-control JSON files were reproduced byte for byte in independent reruns. The two datasets were analysed separately and no joint effect size or pooled animal test was computed.

5 Supplementary exploratory analyses

5.1 Single-tile counterfactual and sequence control

Two geometries, denoted u and o, differed only in whether one lattice tile was blocked. A $u - o$ registered-cell contrast at the focal seam was tested in held-out i, l, and t geometries. Across 46 predictions, the contrast correlated with wall outcomes ($\overline{\rho} = 0.197$) but not open outcomes ($\overline{\rho} = -0.028$), an advantage of 0.225 that was positive in all seven animals. Raw-rate and covariate-adjusted variants gave advantages of 0.243 and 0.225, respectively. A separately eligible near band had mean advantage 0.233 (seven of seven), whereas a far band had mean 0.069 (six of seven); because supports differed, this contrast is descriptive.

The environment sequence was fixed, with u always following o. A same-seam unchanged-state later-minus-earlier placebo was negative (mean -0.042, zero of seven positive), whereas the actual $u - o$ contrast was positive (mean 0.195, seven of seven) and remained positive after nuisance adjustment (0.212). This weakens a generic drift account but cannot remove an environment-specific order effect.

5.2 Direction and two-wall composition

Direction controls did not identify a general orientation rule. Same- versus opposite-signed normals for tangential translation favored the same direction in three of six eligible animals. A normal-axis comparison against opposite-facing or overlapping alternatives was positive in four of seven. A less controlled same- versus opposite-away comparison was positive in all seven, but it also changed strip distance and layout. These analyses support transfer between selected corresponding seams, not position- or direction-independent portability.

Only two complete two-wall factorials were available, both involving the same bit and donut layouts. In five eligible animals, the demeaned additive predictor reached mean correlation 0.153, below an oracle best-single-component predictor at 0.178. The additive advantage was −0.024 and positive in one of five animals. This sparse support provides no positive evidence for linear two-wall composition but cannot establish general non-compositionality.

5.3 Experience and former-barrier endpoints

A transfer-by-experience difference-in-differences estimate was positive under one stored-map specification (mean 0.045, four of four eligible animals), but raw-rate and stricter sensitivity analyses were unstable. Environment order, exposure number, and geometry were not independently varied, so the data do not identify experience-dependent stabilization or innateness.

In an independent neutral-barrier dataset (Blair et al., 2023b), the locked Pearson-active former-barrier center-minus-control endpoint was negative in all leave-one-animal-out summaries (animal mean −0.183; one of six animals positive). A Spearman sensitivity was near zero (mean 0.001; three of six positive). This endpoint showed no persistent local representational scar.

Data and code availability

The boundary-layout source data are available through the archive cited by Lee et al. (2024). Reward-task processed data are available through Figshare+ (Sosa et al., 2025c), with imaging and ROI assets in DANDI 001361 (Sosa et al., 2025b). The independent miniscope data used only for the supplementary former-barrier endpoint are available through Blair et al. (2023a). Boundary analysis code, derived source tables, audit reports, and manuscript sources are available at `https://github.com/asim-v/ca1-wall-update-transfer`. The reward-update analysis code, frozen protocols, machine-readable results, and audit logs are available at `https://github.com/asim-v/ca1-goal-update-transfer`. Neither repository redistributes the original raw recordings, which remain available from their source archives.

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